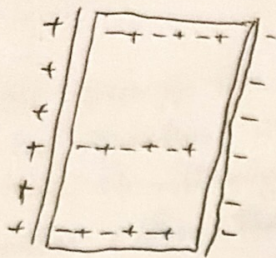


physics and

$H\psi = E\psi$ (one form of Schrödinger wave equation)

Dielectrics



Similar to
Young &
Freedman
figure
24.15
and
24.20

$$\vec{E}_{\text{new}} = \vec{E}_{\text{old}} - \vec{E}_{\text{ind}}$$

$\uparrow$ $\uparrow$ $\uparrow$
 with dielectric without dielectric induced

$$|E_{\text{new}}| < |E_{\text{old}}|$$

K = Dielectric constant

$$K = \frac{E_{\text{old}}}{E_{\text{new}}} = \frac{V_{\text{old}}}{V_{\text{new}}} = \frac{C_{\text{new}}}{C_{\text{old}}}$$

Important string

$$\sigma_i = \sigma \left(1 - \frac{1}{K}\right)$$

Series PPC

$$C = \frac{Q}{\Delta V}$$

$$\Delta V = \frac{Q}{C}$$

$$C = \frac{\epsilon_0 A}{d}$$

$$C =$$